# RROJECT REPORT

# STUDENT MANAGEMENT SYSTEM

## INTRODUCTION:

The provided program is a Student Management System implemented in C++. It allows users to perform various operations such as inserting, searching, updating, deleting, and displaying student records. This report aims to provide a detailed analysis of the program, including its structure, functionality, and potential improvements.

## PROGRAM OVERVIEW:

The program begins by declaring a struct named "student" that represents a student's information, including their ID, name, marks, percentage, and grade. An array of student structures, named 's,' is used to store multiple student records. The program then proceeds to define several functions to perform operations on the student records.

## FUNCTION BREAKDOWN:

a. **`void insert()`:** This function allows users to insert a new student record. It prompts the user to enter the student's ID, name, and marks. Based on the provided marks, the function calculates the percentage and assigns a grade accordingly. The new record is then added to the 's' array.

b**. `void search()`:** This function enables users to search for a student record by their ID. It prompts the user to enter the ID and iterates over the 's' array to find a matching record. If found, it displays the student's name, marks, percentage, and grade. If no record is found, an appropriate message is displayed.

c**. `void update()`:** This function allows users to update an existing student record. Similar to the search function, it prompts the user to enter the ID of the record to be updated. If a matching record is found, the user is prompted to enter the new name and marks. The function then recalculates the percentage and grade based on the updated marks.

d. **`void del()`:** This function allows users to delete a student record. The user is prompted to enter the ID of the record to be deleted. If a matching record is found, all subsequent records are shifted to fill the empty space. The count of records is decremented accordingly.

e**. `void show()`:** This function displays all the existing student records. It iterates over the 's' array and prints the ID, name, marks, percentage, and grade of each student.

f**. `int main()`:** The main function serves as the entry point of the program. It continuously presents a menu of options to the user, allowing them to choose an operation to perform. Based on the selected option, the corresponding function is called. The program runs indefinitely until the user chooses to exit.

**PROGRAM FLOW:**

When the program starts, the main function is called. It displays the main menu and waits for user input. Once the user selects an option, the corresponding function is executed. After completing the chosen operation, the program returns to the main menu. This cycle continues until the user chooses to exit.

**STRENGTH OF THE PROGRAM:**

- The program provides a user-friendly menu-driven interface, making it easy for users to interact with the system.

- It uses a structured approach by utilizing functions to handle specific tasks, improving code readability and maintainability.

- The program incorporates appropriate conditional statements to calculate the percentage and assign grades based on the student's marks.

- The program allows for multiple student records to be stored in an array, enabling efficient management of the data.

**AREA OF IMPROVEMENT:**

Despite the program's functionality, there are several areas that could be improved:

- Error handling: The program lacks comprehensive error handling mechanisms, such as validating user input and handling unexpected scenarios gracefully.

- Memory limitations: The program uses a fixed-size array to store student records, which limits the number of records that can be managed. Implementing dynamic

## PROGRAME

```
#include<iostream>

#include<conio.h>

using namespace std;
```

```
int i=0, found;
struct student {
 int id;
 string name, grade;
  float marks, percentage;
} s[25];
 void insert() {
 system("cls");
 cout<<"\n\n\t\t\t*** Insert Record***";
  cout<<"\n\n Enter student id:";
  cin>>s[i].id;
  cout<<"\n\n Enter Student Name:";
   cin>>s[i].name;
   cout<<"\n\n Enter Student Marks: ";
   cin>>s[i].marks;
   s[i].percentage = s[i].marks/100*100;
   if(s[i].percentage>=75)
   s[i].grade= "A+";
  else if(s[i].percentage>=65)
  s[i].grade= "A";
  else if(s[i].percentage>=55)
  s[i].grade= "B+";
  else if(s[i].percentage>=50)
   s[i].grade= "B";
  else if(s[i].percentage>=45)
  s[i].grade= "C";
  else if(s[i].percentage>=33)
  s[i].grade= "D";
  else
```

```
 s[i].grade= "F";
 i++;
cout<<"\n\n ***Insert Record successfully***";}
void search() {
system("cls");
cout<<"\n\n\t\t\t *** Search record***";
 if(i==0)
 cout<<"\n\n*** No Record Found***";
  else {
  int id;
 found = 0;
 cout<<"\n\n Enter Student id:";
 cin>>id;
 for(int a=0;a<i;a++) {
 if( id==s[a].id) {
 cout<<"\n\n Student Name:"<<s[a].name;
 cout<<"\n\n Student Marks:"<<s[a].marks;
 cout<<"\n\n Student percentage:"<<s[a].percentage<<"%";
 cout<<"\n\n Grade"<<s[a].grade;
 found++;}
 }
    }
 if( found==0){
 cout<<"\n\n*** Student id Not Found***";}
}
void update() {
system("cls");
cout<<"\n\n\t\t\t *** Update record***";
if(i==0)
```

```
cout<<"\n\n*** No Record Found***";
else {
int id;
found = 0;
cout<<"\n\n Enter Student id:";
 cin>>id;
 for(int a=0;a<i;a++) {
 if( id==s[a].id) {
 cout<<"\n\n Enter Student Name:";
 cin>>s[a].name;
 cout<<"\n\n Enter Student Marks: ";
 cin>>s[a].marks;
 s[a].percentage = s[a].marks/100*100;
 if(s[a].percentage>=85)
 s[a].grade= "A+";
 else if(s[a].percentage>=75)
 s[a].grade= "A";
 else if(s[a].percentage>=65)
 s[a].grade= "B+";
  else if(s[a].percentage>=55)
  s[a].grade= "B";
  else if(s[a].percentage>=50)
  s[a].grade= "C";
  else if(s[a].percentage>=45)
  s[a].grade= "D";
  else if(s[a].percentage>=33)
  s[a].grade= "E";
  else
  s[a].grade= "F";
```

```
cout<<"\n\n*** Update Record Successfully***";
found++;}
 }
 if(found==0)
 cout<<"\n\n*** Student Id Not Found***";}
}
void del() {
 system("cls");
 cout<<"\n\n\t\t\t *** Delete record***";
  if(i==0)
  cout<<"\n\n*** No Record Found***";
  else {
   int id;
   found = 0;
   cout<<"\n\n Enter Student id:";
   cin>>id;
   for(int a=0;a<i;a++) {
   if(id==s[a].id) {
    for(int m=a;m<i-1;m++) {
     s[m].id = s[m+1].id;
     s[m].name = s[m+1].name;
      s[m].marks = s[m+1].marks;
      s[m].percentage = s[m+1].percentage;
     s[m].grade = s[m+1].grade;}
     cout<<"\n\n Delete Record Successfully";
     found++;
     i--;
      break;}
      }
```

```
if(found==0) {
cout<<"\n\n*** Student Id Not Found***";}
 }
}
void show() {
system("cls");
cout<<"\n\n\t\t\t *** Show All record***";
if(i==0)
cout<<"\n\n*** No Record Found***";
else {
for(int a=0;a<i;a++) {
cout<<"\n\n Student Id: "<<s[a].id;
cout<<"\n\n Name: "<<s[a].name;
cout<<"\n\n Marks: "<<s[a].marks;
 cout<<"\n\n Percentage: "<<s[a].percentage<<"%";
 cout<<"\n\n Grade: "<<s[a].grade;
 cout<<"\n\n****************************";}
 }
 }
int main() {
int choice;
 while (true) {
  system("cls");
   cout<<"\n\n\t\t\t***STUDENT MANAGEMENT SYSTEM***";
    cout<<"\n\n 1. Insert Record";
    cout<<"\n\n 2. Search Record";
    cout<<"\n\n 3. Update Record";
     cout<<"\n\n 4. Delete Record";
     cout<<"\n\n 5. Show Record";
```

```
cout<<"\n\n 6. Exit";
cout<<"\n\n Enter Your Choice->";
cin>>choice;
switch(choice) {
case 1:
insert();
break;
case 2:
search();
break;
case 3:
update();
break;
case 4:
del();
 break;
 case 5:
 show();
 break;
 case 6:
 exit (0);
 default:
 cout<<"\n\n*** Invalid Choice";}
 getch();}
return 0;}
```

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